

Mugdha Pandya

PhD Student @ CDT for Designing Responsible NLP
School of Informatics, University of Edinburgh

Research Statement

My research sits at the intersection of hate speech detection and cultural representation in NLP. Current systems are built predominantly on Western-centric data and treat meaning as a fixed property of text, but what counts as harmful language and who it targets depend deeply on cultural context and audience interpretation. My PhD investigates how geo-cultural biases manifest in hate speech datasets and models, and how implicit and coded forms of harmful language can be better captured computationally. I draw on interdisciplinary methods across NLP, social science, and linguistics to develop approaches that preserve diverse cultural understandings of harm in both data collection and modelling.

Research Interests

Natural Language Processing, Hate Speech Detection, Implicit and Coded Language, Geo-cultural Bias, Fairness and Representation in NLP, Annotator Disagreement

Educational Qualifications

Programme	University	Institute	Year
M.Sc Artificial Intelligence	University of Edinburgh	School of Informatics, University of Edinburgh	2020
B.Tech, Computer Science and Engineering	VIT, India	School of Computer Science, Vellore Institute of Technology, Vellore	2018

Publications

Hostility Detection in UK Politics: A dataset on online Abuse Targeting MPs

Pandya, M., Jin, M., Bontcheva, K., & Maynard, D. (2025, August). Hostility detection in UK politics: A dataset on online abuse targeting MPs. In Proceedings of the The 9th Workshop on Online Abuse and Harms (WOAH) (pp. 254-266).

Dimensions of Online Conflict: Towards Modeling Agonism

Canute, M., Jin, M., Holtzclaw, H., Lusoli, A., Adams, P., Pandya, M., ... & Chun, W. H. K. (2023, December). Dimensions of Online Conflict: Towards Modeling Agonism. In Findings of the Association for Computational Linguistics: EMNLP 2023 (pp. 12194-12209).

Shefnlp at SemEval-2023 Task 10: Compute-Efficient Category Adapters

Thomas Pickard, Tyler Loakman, and Mugdha Pandya. 2023. shefnlp at SemEval-2023 Task 10: Compute-Efficient Category Adapters. In Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023), pages 1069–1075, Toronto, Canada. Association for Computational Linguistics.

Rana Ayyub: Targeted online violence at the intersection of misogyny and Islamophobia.

Posetti, J., Bontcheva, K., Zaffar, H., Shabbir, N., Maynard, D. and Pandya, M. (2023). Rana Ayyub: Targeted online violence at the intersection of misogyny and Islamophobia.

Alexa, Google, Siri: What are Your Pronouns? Gender and Anthropomorphism in the Design and Perception of Conversational Assistants

Abercrombie, G., Curry, A. C., Pandya, M., & Rieser, V. (2021, August). Alexa, Google, Siri: What are Your Pronouns? Gender and Anthropomorphism in the Design and Perception of Conversational Assistants. In Proceedings of the 3rd Workshop on Gender Bias in Natural Language Processing (pp. 24-33).

Intrinsic Bias Metrics Do Not Correlate with Application Bias

Goldfarb-Tarrant, S., Marchant, R., Sánchez, R. M., Pandya, M., & Lopez, A. (2021, August). Intrinsic Bias Metrics Do Not Correlate with Application Bias. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (pp. 1926-1940).

Job Experience

Research Associate

November 2021 – June 2025

Location – University of Sheffield, Sheffield

- Responsible AI for Inclusive, Democratic Societies: A cross-disciplinary approach to detecting and countering abusive language online
 - Developing computational methods and analysing social media content for intersectional hate speech detection as a part of the GATE team. In particular, investigating bias in natural language processing methods for use cases in analysing online engagement with and hate speech towards women journalists and UK MPs. It includes the creation of benchmark datasets and machine-learning models.
 - Investigating bias in hate speech and sexism detection datasets due to lack of minority representation through label aggregation and the consequent bias amplification in machine-learning models trained on such data. Also, improving the detection of nuanced and subtle forms of hate speech and sexism through minority annotator representation.
- Advancing FrontTier Research In the Arts and hUMANities (ATRIUM)
 - Developing NLP tools for processing archaeological texts, specifically focused on Named Entity Recognition. Also, working on text extraction and summarisation methods for long-form archaeological investigation reports.

Research Assistant

September 2020 – September 2021

Location – Heriot-Watt University, Edinburgh

Designing Conversational Assistants to Reduce Gender Bias under the guidance of Dr Verena Rieser at The Interaction Lab. Designing and developing alternative conversational personas that are less likely to perpetuate bias by studying conversational gendering (expressed through voice, content, and style). Worked with state-of-the-art data-efficient NLP deep learning tools for generating dialogue responses consistent with a given persona.

Project Experience

How Geometry of Word Embeddings Affects Downstream Applications

May 2020 – August 2020

Location – University of Edinburgh, Edinburgh

M.Sc. thesis project under Dr Adam Lopez's guidance involving evaluation of whether modification of word embeddings to remove bias is sufficient to improve fairness in downstream applications. This was done by correlating intrinsic bias measures in word embedding vector space with extrinsic measures of bias found in standard downstream NLP applications.

Content-based search for audio lectures in Indic languages

July 2018 – March 2019

Location – Indian Institute of Technology Bombay, Mumbai

Worked under the guidance of Dr Ganesh Ramakrishnan on a project involving constructing a speech-to-text system that caters to Indian English and Indic languages along with word alignments, an automatic tag phrase generation system with the timing for parts of videos, a topic modelling system for audio lectures and a speech-cum-text based query system. An automatic speech recogniser for Gujarati and Hindi languages using the Kaldi tool and a content-based search engine for audio files using Elasticsearch was built.

Analyses & Segmentation of Code-Switched Speech in Indian Languages

December 2017 – May 2018

Location – Indian Institute of Technology Bombay, Mumbai

Analysed code-switched speech involving English and Hindi to find lexical and acoustic features that can help in language differentiation under the guidance of Dr Preeti Rao. Built a basic language identifier system that automatically segments a speech at pauses and classifies each segment into Hindi and English using these features.

Location – Tata Institute of Fundamental Research, Mumbai

Built a system for classifying Hindustani music into different raags under the guidance of Dr Samudravijaya K. The system used techniques from statistical pattern recognition and natural language processing. Further work was done using HMM and conditional probability models for better results.

Skills

Programming Languages - Python, C, C++, Java, PHP, SQL, JavaScript, HTML, Praat, Shell script

NLP and ML Libraries - HuggingFace, PyTorch, scikit-learn, NLTK, Gensim, SpaCy, NumPy, Pandas, SciPy

Visualisation - Matplotlib, Seaborn, Plotly

Other Tools - GitHub, Qualtrics, Prolific, Elasticsearch, Google Colab, MATLAB, Flask, OpenFst, OpenCV libraries, Google Cloud Speech API, Kaldi, HTK Speech Recognition Toolkit